

Depth of Closure (DoC) Calculator & Sensitivity Analysis

Sediment Transport Limits for Coastal Engineering

1. Abstract

This repository contains computational tools designed to calculate the **Depth of Closure (DoC)**—the theoretical depth at which sediment exchange between the nearshore and offshore becomes negligible. The software implements the most widely accepted empirical formulations from coastal engineering literature: **Birkemeier (1985)**, **Hallermeier (1981/1983)**, and **Houston (1995)**.

A distinguishing feature of this tool is its built-in **Sensitivity Analysis**. Coastal data is often subject to uncertainty; therefore, this calculator not only computes the DoC for the base case but automatically simulates five additional scenarios (Low/High Energy, Short/Long Period, and Worst Case). This allows engineers to understand the variability of the closure depth relative to wave climate fluctuations.

The tools provide specific recommendations for three distinct engineering applications:

1. **Beach Nourishment** (Active Profile Toe).
2. **Hard Structures** (Scour Protection foundations).
3. **Dredged Material Disposal** (Stable Mound placement).

2. Theoretical Framework & Methodology

The software derives statistical wave parameters from annual climate data and applies them to empirical depth limits.

2.1 Wave Statistics Derivation

The inputs require the Annual Mean Significant Wave Height ($H_{s,mean}$) and the Annual Maximum Significant Wave Height ($H_{s,max}$). The software assumes a statistical distribution to derive the **Effective Wave Height** (H_e), which represents the extreme tail of the distribution (approx. 0.137% exceedance) required for Inner DoC calculations.

Standard Deviation Estimation (σ):

$$\sigma \approx \frac{H_{s,max} - H_{s,mean}}{5.6}$$

Effective Wave Height (H_e):

$$H_e = H_{s,mean} + 5.6\sigma$$

Median Wave Height (H_m): Used for Outer DoC calculations:

$$H_m = H_{s,mean} - 0.3\sigma$$

2.2 Empirical Depth Models

The calculator implements four distinct formulations to determine the depth of closure (h_c).

1. Birkemeier (1985) - Field Adjusted Inner Limit: A modification of the Hallermeier equation based on extensive field measurements at the CERC Field Research Facility. It generally predicts a shallower depth, optimizing sand volume for nourishment.

$$h_c = 1.75H_e - 57.9 \left(\frac{H_e^2}{gT_e^2} \right)$$

2. Hallermeier (1981) - Analytical Inner Limit: Based on the critical Froude number for intense bed agitation. This provides a conservative (deeper) limit, recommended for structural foundations to prevent scour.

$$h_c = 2.28H_e - 68.5 \left(\frac{H_e^2}{gT_e^2} \right)$$

3. Houston (1995) - Simplified Parameterization: A rapid estimation based solely on the mean wave height.

$$h_c = 6.75H_{s,mean}$$

4. Hallermeier (1983) - Outer Limit (Sediment Entrainment): Defines the seaward limit of the "Shoal Zone" where typical waves can no longer agitate the specific sediment size (D_{50}). Used for siting dredged material disposal mounds.

$$h_c = 0.018H_mT_e\sqrt{\frac{g}{(s-1)D_{50}}}$$

Where s is the specific gravity of the sediment (typically 2.65).

3. Repository Contents & Compilation

This repository includes four implementation variations to suit different engineering workflows.

3.1 Python Script (depth_of_closure.py)

A rapid prototyping script ideal for quick checks or integration into larger pipelines.

- **Dependencies:** Python 3.x (Standard libraries: `math`, `sys`, `argparse`, `pandas`, `numpy`).
- **Execution:**
`python3 depth_of_closure.py`

3.2 Jupyter Notebook (depth_of_closure.ipynb)

An interactive notebook version of the calculation engine. This format is ideal for educational purposes, step-by-step code walkthroughs, and visualizing the sensitivity analysis data.

- **Dependencies:** Jupyter Lab/Notebook, Pandas, NumPy.
- **Execution:**
`jupyter notebook depth_of_closure.ipynb`

3.3 C++ CLI (depth_of_closure_cli.cpp)

A high-performance command-line tool designed for batch processing. It performs the full sensitivity analysis and exports a formatted text report.

- **Compilation (GCC/MinGW):**
`g++ -O3 -march=native -std=c++17 -Wall -Wextra \`
`-static -static-libgcc -static-libstdc++ \`
`-o depth_of_closure_cli depth_of_closure_cli.cpp`
- **Usage (Interactive):** `./depth_of_closure_cli`
- **Usage (Arguments):**
`./depth_of_closure_cli [Hs_Mean] [Te] [Hs_Max] [D50_mm] [SG]`
Example:
`./depth_of_closure_cli 2.25 9.0 7.7 0.50 2.65`

3.4 C++ GUI (depth_of_closure_gui.cpp)

A standalone native Windows application using the Win32 API. It provides a visual interface for inputting wave parameters and viewing the generated "Design Decision Matrix."

- **Compilation (MinGW on Windows):**
`g++ -O3 -march=native -std=c++17 -unicode \`
`depth_of_closure_gui.cpp -o depth_of_closure_gui \`
`-mwindows -static -static-libgcc -static-libstdc++`
- **Features:**
 - Form-based inputs with validation.
 - Real-time generation of the Technical Report.
 - Automatic file export to `output.txt`.

4. Input Parameter Definitions

Parameter	Symbol	Unit	Description
Annual Mean Hs	$H_{s,mean}$	meters	The yearly average significant wave height.
Mean Period	T_e	seconds	The mean wave period associated with the wave clim
Annual Max Hs	$H_{s,max}$	meters	The maximum significant wave height recorded in a t
Sediment Size	D_{50}	mm	The median grain diameter of the native beach sedim
Specific Gravity	s	-	Relative density of sediment (Default Quartz = 2.65)

5. Output Report Explanation

The software generates a detailed technical report. Key sections include:

1. **Executive Summary:** Summarizes the input site conditions and calculated statistical parameters (H_e, H_m, σ).
 2. **Design Decision Matrix:** Recommends specific depth values based on the intended application:
 - **[A] Beach Nourishment:** Uses Birkemeier (Shallower).
 - **[B] Hard Structures:** Uses Hallermeier Inner (Deeper/Safer).
 - **[C] Disposal:** Uses Hallermeier Outer (Deepest/Most Stable).
 3. **Sensitivity Analysis Table:** A matrix displaying results for different wave scenarios.
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6. Bibliographic References

1. **Birkemeier, W.A. (1985).** "Field data on Seaward Limit of Profile Change." *Journal of Waterway, Port, Coastal and Ocean Engineering*, 111(3), 598-602.
 2. **Hallermeier, R.J. (1981).** "A profile zonation for seasonal sand beaches from wave climate." *Coastal Engineering*, 4, 253-277.
 3. **Hallermeier, R.J. (1983).** "Sand transport limits in coastal structure design." *Proceedings of Coastal Structures '83*, ASCE.
 4. **Houston, J.R. (1995).** "Beach Nourishment." *Coastal Engineering Technical Note*, CETN-II-32, USACE.
 5. **USACE (2002).** *Coastal Engineering Manual (CEM)*, Part III.
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